

Harshavarthan Venkatesan

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● ABOUT ME

Artificial Intelligence and Data Science undergraduate with hands-on experience in deep learning, computer vision, embedded systems and full-stack development. Currently Student Lead for Technology and Research at Microcosm, leading a Harvard-funded computer vision project on elephant behaviour and human-elephant conflict mitigation. Former embedded engineer with Team Sea Sakthi at the Monaco Energy Boat Challenge. Interested in applying AI to real-world problems.

● EDUCATION & TRAINING

09/09/2023 - Current - COIMBATORE, INDIA

BACHELOR OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE AND DATA SCIENCE- KUMARAGURU COLLEGE OF TECHNOLOGY

AI & Machine Learning, Applied Machine Learning, Neural Networks & Deep Learning, Reinforcement Learning, Data Science, Data Mining & Modelling, EDA & Visualisation, Cloud Architecture, Industrial IoT, Business Intelligence.

Field(s) of study: Artificial Intelligence and Data Science | **Final grade:** 7.4/10 | **Level in EQF:** 6 | **Number of credits:** 162 | **Thesis:** Elephant Behavioural Analysis and Mitigation System – AI-Powered Computer Vision for Non-Invasive Elephant Behaviour Monitoring and Human–Elephant Conflict Mitigation (Supported by a USD 11,000 Harvard University grant)

01/01/2026 - 30/06/2026 - COIMBATORE, INDIA

GRADUATE CERTIFICATE IN INDUSTRIAL DIGITAL TECHNOLOGIES- FORGE INNOVATION & VENTURES

Natural Language Processing, Computer Vision Techniques, Startup Fundamentals, Applied Design Thinking, Prototype Development, Industrial IoT

Final grade: Semester VI GPA – 9.1/10

01/06/2021 - 05/04/2023 - COIMBATORE, INDIA

HIGHER SECONDARY CERTIFICATE (HSC) – TAMIL NADU STATE BOARD- LISIEUX MATRICULATION HIGHER SECONDARY SCHOOL

Computer Science, Physics, Chemistry, Mathematics, English, Tamil

Field(s) of study: Computer Science | **Final grade:** 91%

● WORK EXPERIENCE

01/07/2026 - CURRENT - COIMBATORE, INDIA

STUDENT LEAD – TECHNOLOGY AND RESEARCH PROJECTSMICROCOSM, KUMARAGURU COLLEGE OF TECHNOLOGY

Lead, mentor and contribute hands-on to student research projects, including the Harvard-funded elephant, diary-based computer vision, campus carbon-footprint and dashboard projects.

Website: <https://microcosm.kct.ac.in/>

EMBEDDED ENGINEERTEAM SEA SAKTHI, KUMARAGURU COLLEGE OF TECHNOLOGY

Contributed to the embedded systems team developing the Yali 4.0 electric catamaran for the 12th Monaco Energy Boat Challenge and an autonomous river-cleaning boat, working on cloud telemetry, FPV-AR and real-time monitoring, and independently built the team website.

Website: <https://seasakthi.com/>

PROJECTS

01/10/2024 - 30/09/2025

Yali 4.0 - TeamSeaSakthi Contributed to the embedded systems of an electric catamaran and an autonomous river-cleaning boat, working with microcontrollers, sensors, communication protocols, FPV-AR, cloud telemetry and dashboards, and represented the team at the 12th Monaco Energy Boat Challenge (8th place, Communication and Townhall Cup).

01/05/2026 - 30/04/2027

Elephant Behavioural Analysis and Mitigation System Lead a 13-member team on a USD 11,000 Harvard Mittal Institute grant project, contributing to models for elephant classification, identification, herd, feeding, gait and aggression analysis, and conflict-risk detection using YOLO, ByteTrack, CNN-XGBoost, MobileNetV3, DeepLabCut and 3D ResNet-18, integrated into an AI monitoring dashboard.

01/06/2026 - 15/10/2026

Sustainability Intelligent Dashboard - Kumaraguru Institutions Co-developed a campus sustainability platform (Python, FastAPI, PostgreSQL) as lead of a 4-member team, covering energy, GHG, water and waste data with Scope 1 and 2 GHG calculations aligned with ISO 14064-1:2018.

Link <https://sustainability.kct.ac.in>

17/07/2026 - 17/10/2026

K-geo portal Built a Next.js and Supabase portal while leading a 3-member team, featuring Gemini-powered university recommendations, role-based access control and automated PDF reports.

10/07/2026 - CURRENT

Sugarcane Research Analytical Platform Developed and led a full-stack React and Supabase analytics platform for a Microcosm and Environmental Defense Fund field study (3 plots, 14 treatments), with growth, fertiliser-response and heatmap analytics.

SKILLS

computer vision | deep learning | machine learning | embedded systems | Internet of Things | Python (computer programming) | JavaScript | Embedded C | TypeScript | CSS | SQL | PostgreSQL | Full-Stack Development | ICT communications protocols | ICT system integration | sensors | perform data analysis | project management | manage teamwork | communicate with stakeholders | think innovatively | think analytically | evaluate project plans

LANGUAGE SKILLS

Mother tongue(s): **TAMIL**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C1	B2	B2	B2
GERMAN	A2	A2	A2	A2	A2

HINDI	B1	B1	B1	B1	B1
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● HONOURS AND AWARDS

05/07/2025 Yacht Club de Monaco

Certificate of Achievement – Energy Class Team Sea Sakthi, Kumaraguru College of Technology – 8th Place and Communication and Townhall cup at the 12th Monaco Energy Boat Challenge (1–5 July 2025).

Link https://harshavarthan-umber.vercel.app/MEBC_Certificate.pdf

15/08/2025 Kumaraguru Institutions

Certificate of Recognition Recognition for contributions to Team Sea Sakthi, which represented Kumaraguru College of Technology at the Monaco Energy Boat Challenge 2025.

29/09/2026 Microcosm

Appreciation letter Appreciated for taking initiative and leading technical and research projects at Microcosm, demonstrating dedication through mentoring, team upscaling, innovation, and a strong commitment to driving impactful outcomes.

Link https://harshavarthan-umber.vercel.app/Appreciation_Letter_Microcosm.pdf

15/09/2026 Kumaraguru Global Engagement Office

Appreciation letter Recognized for taking the initiative and leading the development of a portal to track students pursuing opportunities abroad. Appreciated for demonstrating responsibility, ownership, and innovation in supporting the Global Engagement Office.

Link https://harshavarthan-umber.vercel.app/Appreciation_Letter_Kgeo.pdf

● CERTIFICATIONS

15/07/2026 FORGE Innovation & Ventures

Graduate Certificate in Industrial Digital Technologies – PROTOSEM Innovation Engineers

Project Based - Completed the PROTOSEM Innovation Engineers programme (Cohort 1.17), applying technical knowledge through hands-on project development, prototyping, and innovation to solve real-world industrial problems.

04/09/2026 DeepLearning.AI

Deep Learning Specialization

Online - Completed the Deep Learning Specialization covering neural networks, CNNs, sequence models, and practical deep learning applications.

17/03/2025 University of Illinois Urbana-Champaign

Data Visualization

Online - Completed a course covering data visualization techniques for analyzing and communicating data effectively using visual methods.

28/05/2024 IBM

Introduction to Cybersecurity Tools & Cyberattacks

Completed a course covering fundamental cybersecurity tools, common cyberattacks, and essential security concepts.